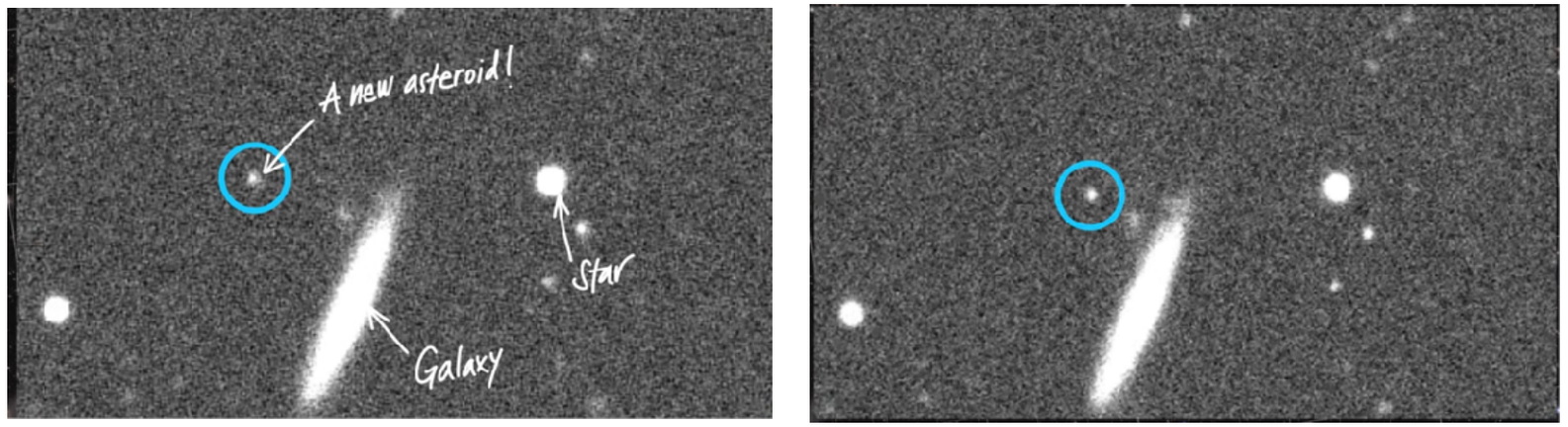
LSST Solar System data access

Pedro Bernardinelli
University of São Paulo &
Rubin Solar System
Processing team

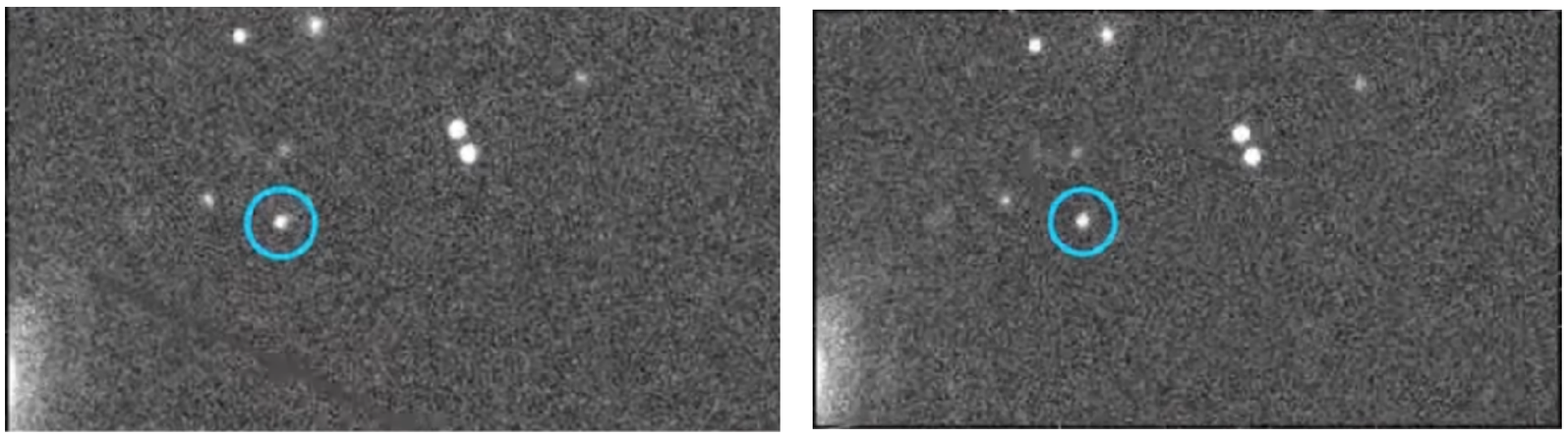
Rubin Observatory's Solar System Processing (SSP) algorithms search for solar system objects by looking for sources that appear to be moving over several nights

Observation 1 | Observation 2 (~30 minutes later)

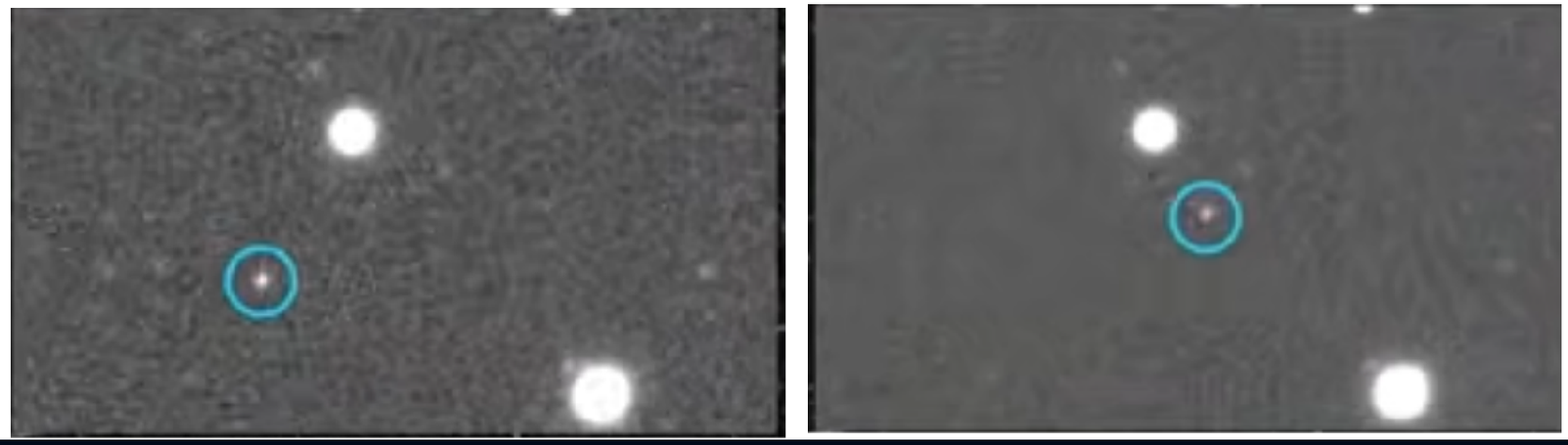
Night 1



Night 2



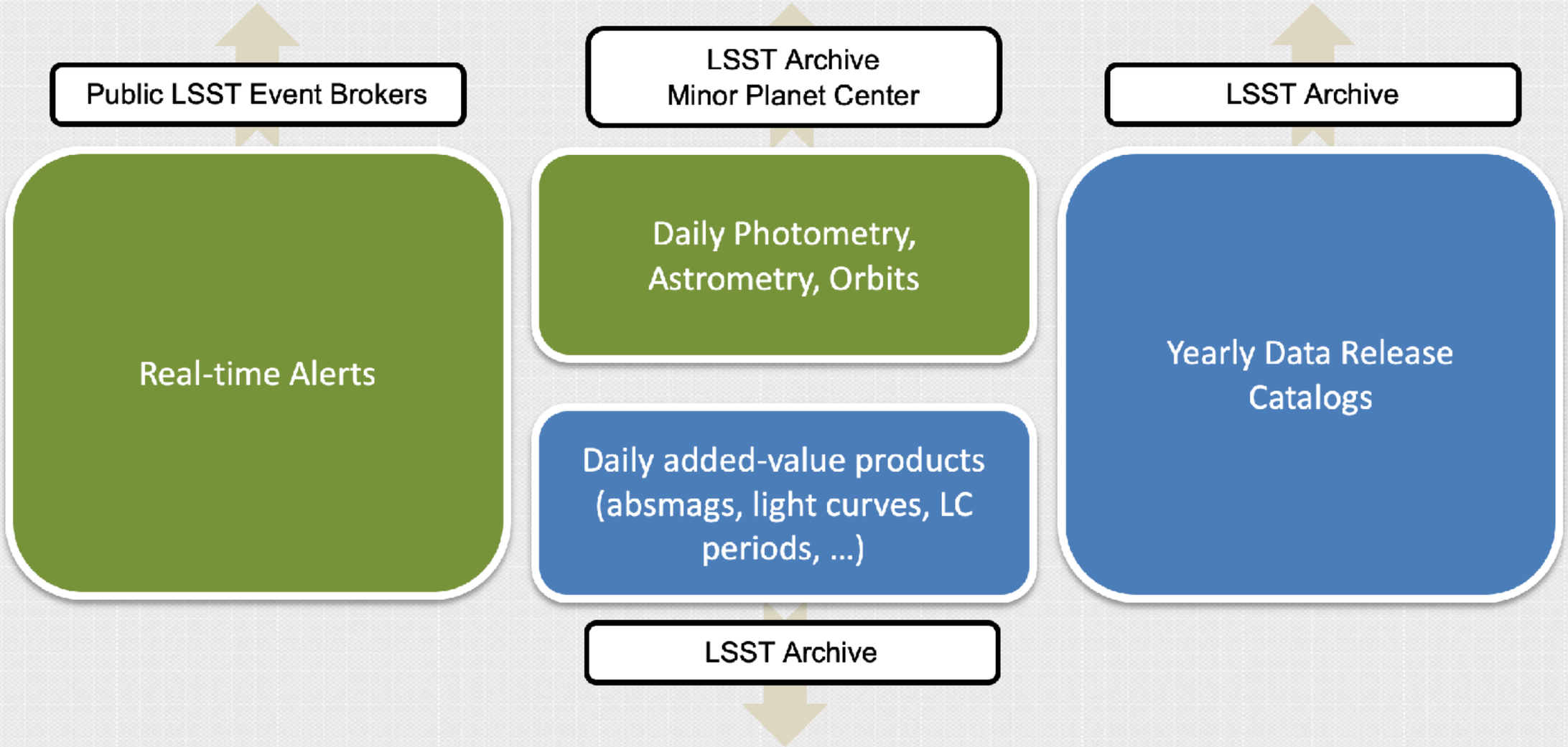
Night 3



Graphics Credit: NSF–DOE Rubin Observatory

Slide credits: Meg Schwamb (QUB)

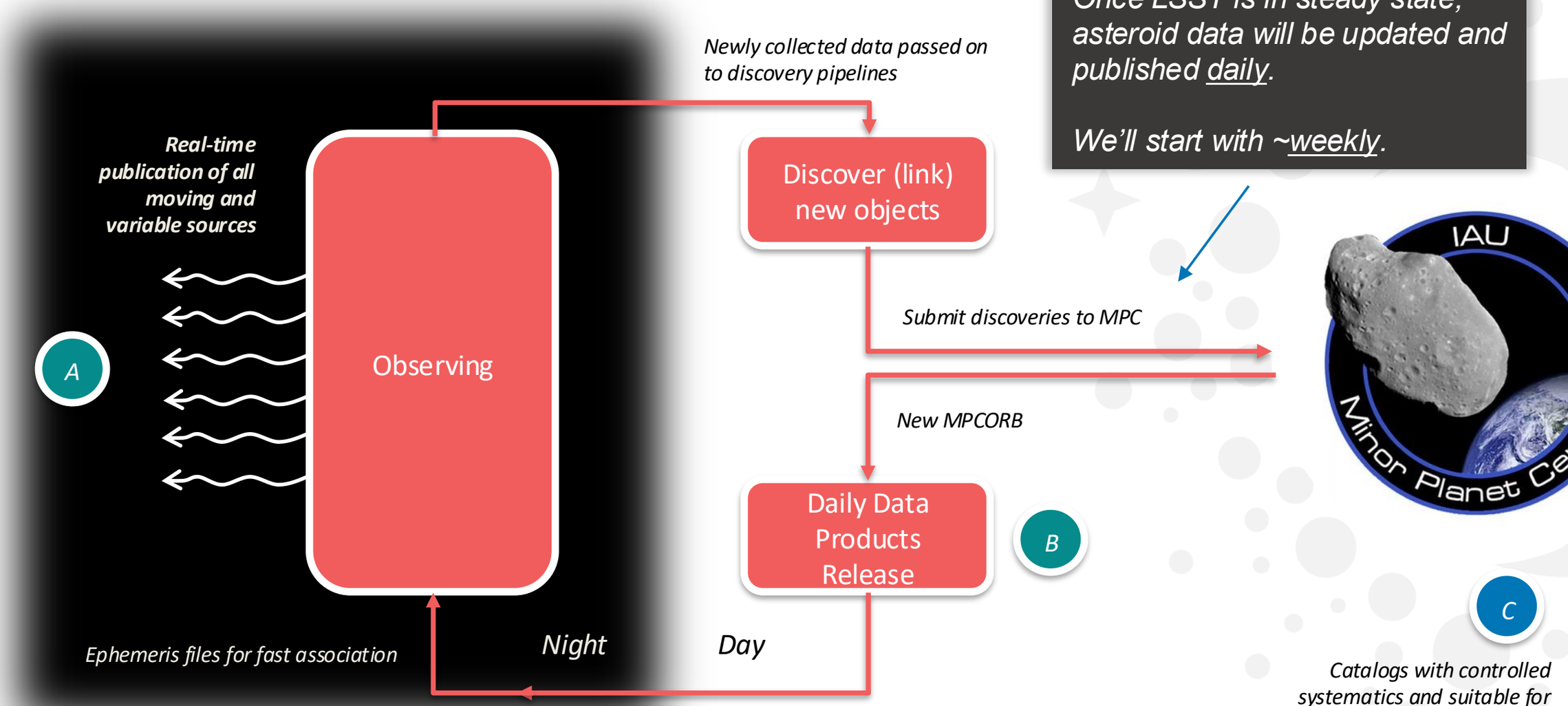
Obtaining the LSST Solar System Data Products



Graphics Credit: Mario Juri

Slide credits: Meg Schwamb (QUB)

LSST Nightly/Daily Processing Loop



Graphics Credit: Mario Jurić

Slide credits: Meg Schwamb (QUB)

The International Astronomical Union
Minor Planet Center

OBSERVERS DATA NEWSLETTER STATUS

2025 MT343

Initial reported observation by Simonyi Survey Telescope, Rubin Observatory on 2025-06-21.

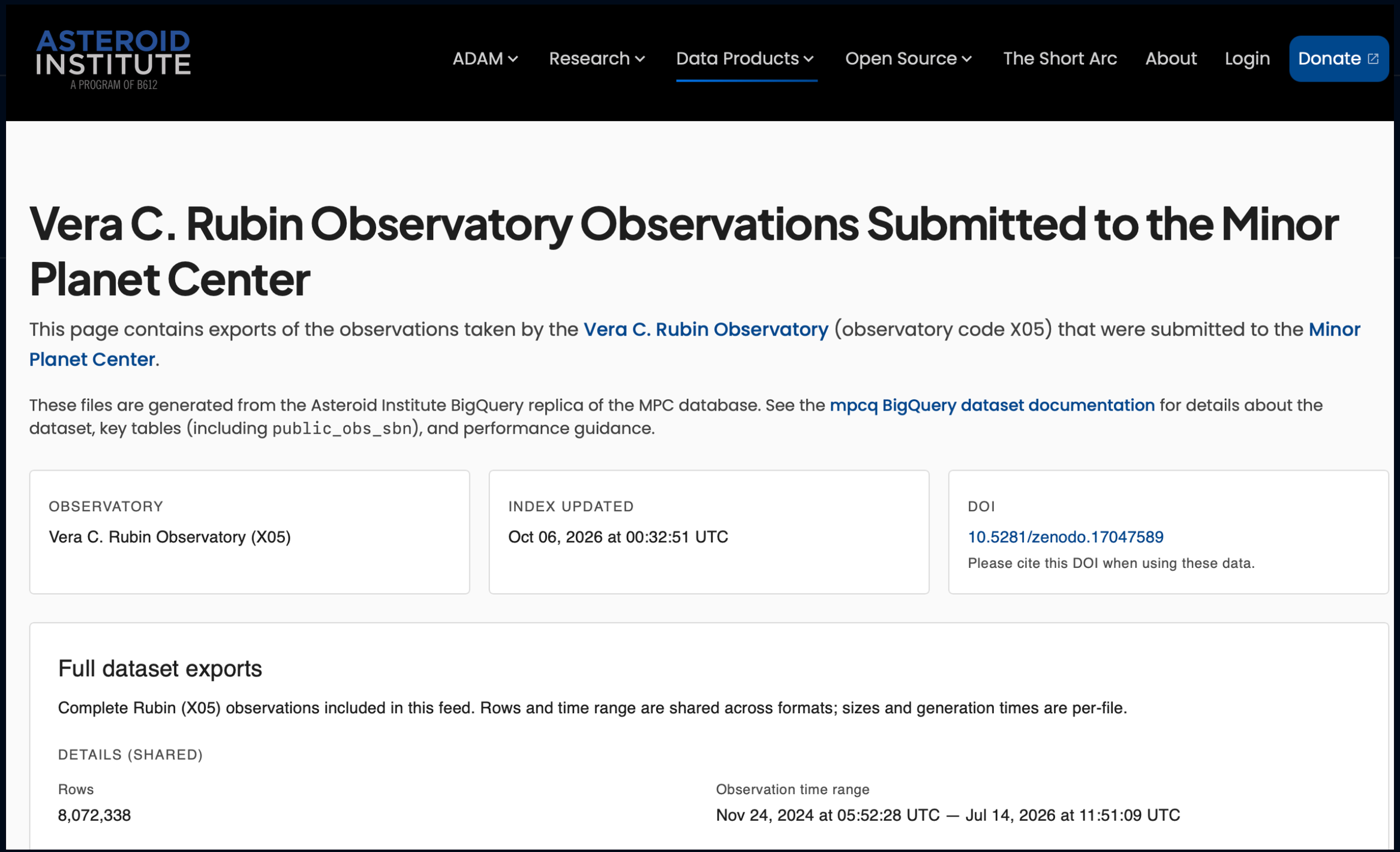
(Discoverer will be defined when the object is numbered. See this note on how discoverers are determined.)

Orbit

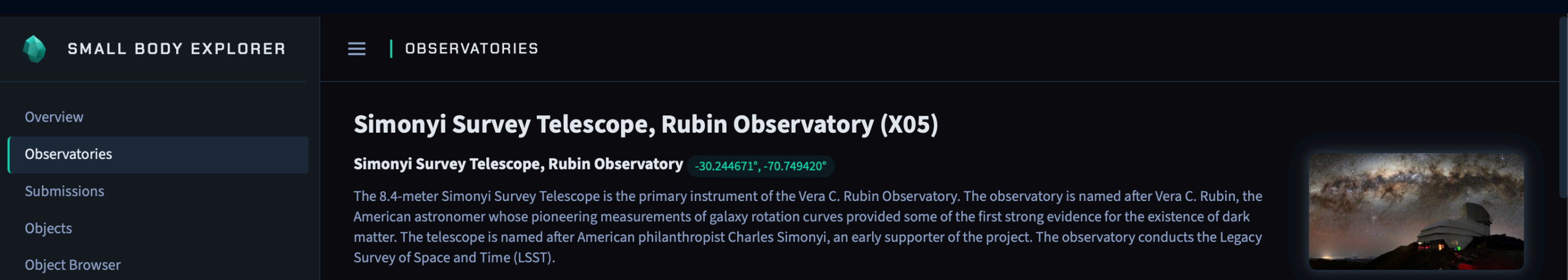
Orbit type: Main Belt

- Example object initially reported by LSST
- MPC has Rubin data (X05), but also data from other observatories (eg F51)

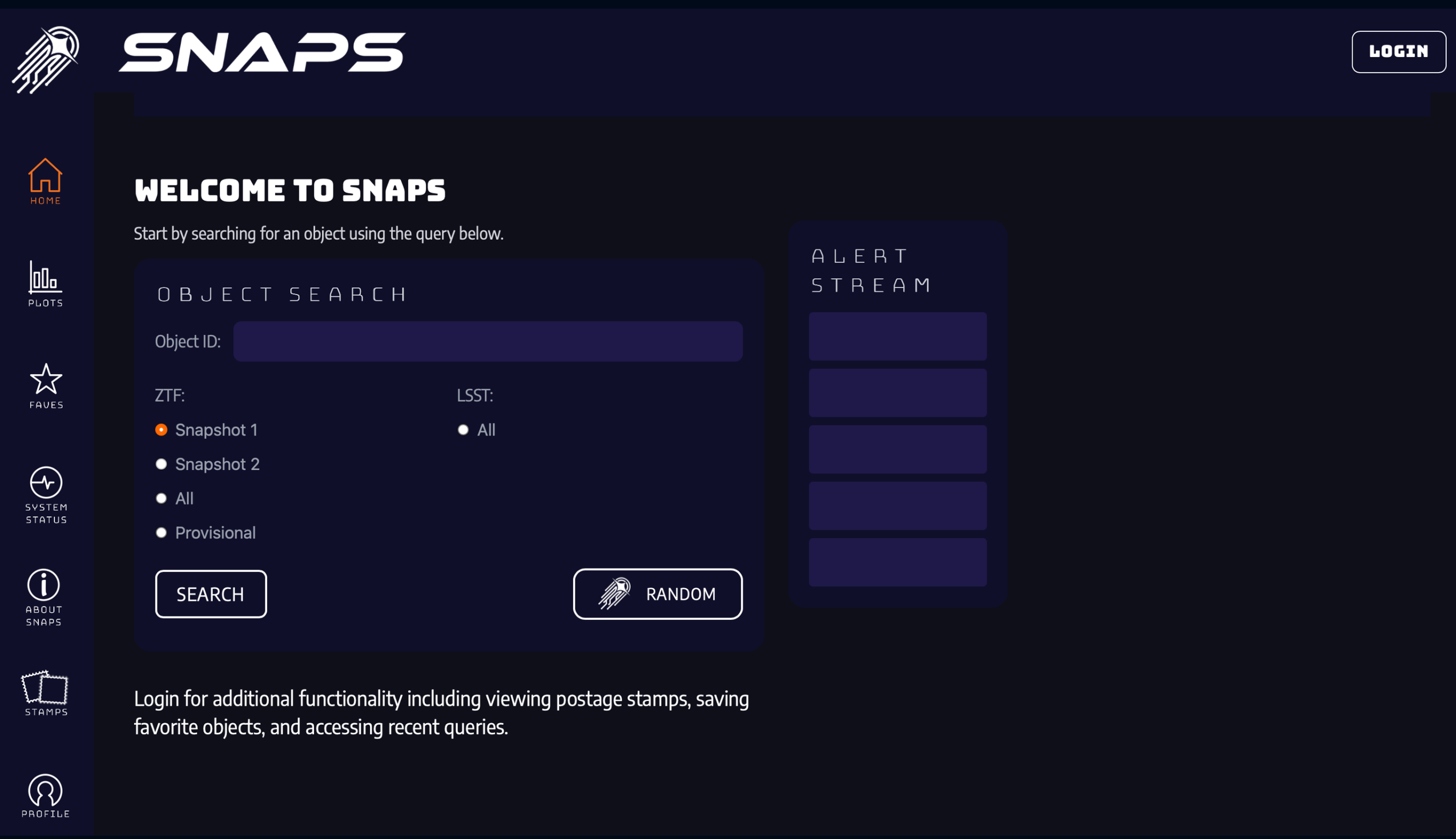
2024 05 09.307947	12 15 57.664	−24 24 08.49	21.51 w	F51 – Pan-STARRS 1, Haleakala	MPS 2549096
2024 05 09.311182	12 15 57.572	−24 24 07.29	21.50 w	F51 – Pan-STARRS 1, Haleakala	MPS 2549096
2025 06 21.290163	21 53 15.821	−14 44 27.07	22.11 i	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 06 21.314677	21 53 15.637	−14 44 21.98	22.11 z	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 07 21.106820	21 39 39.855	−13 36 28.06	21.79 r	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 07 21.140360	21 39 38.292	−13 36 25.61	21.62 i	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 07.365697	21 24 40.848	−13 21 47.44	21.16 i	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 07.380419	21 24 40.029	−13 21 46.75	21.34 r	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 14.338361	21 18 15.288	−13 17 46.85	21.35 r	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 14.350436	21 18 14.616	−13 17 46.34	21.13 i	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 17.204297	21 15 40.182	−13 16 10.66	22.10 g	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 17.205838	21 15 40.095	−13 16 10.58	22.02 g	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 08 18.406813	21 14 36.125	−13 15 33.27	21.53 w	F51 – Pan-STARRS 1, Haleakala	MPS 2549096
2025 08 18.419814	21 14 35.410	−13 15 32.86	21.55 w	F51 – Pan-STARRS 1, Haleakala	MPS 2549096
2025 08 18.432655	21 14 34.716	−13 15 32.56	21.50 w	F51 – Pan-STARRS 1, Haleakala	MPS 2549096
2025 09 03.237867	21 02 15.177	−13 04 04.57	21.44 z	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 09 03.246022	21 02 14.860	−13 04 04.11	21.67 i	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096
2025 09 03.254251	21 02 14.529	−13 04 03.61	21.69 i	X05 – Simonyi Survey Telescope, Rubin Observatory	MPS 2549096



- The Asteroid Institute is compiling and redistributing all Rubin data that is available in the MPC
  - Link: https://b612.ai/rubin-mpc-downloads/



- How do you keep track of Rubin discoveries?
  - Well, internally we had this problem too…
- Tool initially meant for internal use only, but quite widespread now: https://sbx.dirac.dev/station/X05



- SNAPS: the only broker dedicated for Solar System objects, currently in late stages of development at NAU
  - https://snaps.nau.edu/home